SC12 - Revision E purchasing BOM

120 fitted parts: 111 factory SMT + 9 user-fitted through-hole parts. Three loose caps extra. Parts purchasing estimate: $35.14/board at 10 boards; PCB, assembly, attrition, freight and tax extra. See cost_estimate.csv. Unbuilt prototype.

Qty	MPN / supplier part	References	Value / package	USD first tier
1	0451003.MRL / C13982	F1	3A FAST / Fuse_Littelfuse-NANO2-451_453	0.2556
5	0805W8F1001T5E / C17513	R4, R30, R110, R210, R310	1k / R_0805_2012Metric	0.021
10	0805W8F1002T5E / C17414	R3, R21, R22, R40, R41, R105, R205, R305, R63, R52	10k / R_0805_2012Metric	0.039
8	0805W8F1003T5E / C149504	R50, R65, R111, R112, R211, R212, R311, R312	100k / R_0805_2012Metric	0.048
3	1824420000 / 281-2020-ND	J1, J2, J3	2P PUSH 3.5mm / Weidmuller_LSF_90_2P	4.89
3	1824430000 / 281-2021-ND	J4, J5, J6	3P PUSH 3.5mm / Weidmuller_LSF_90_3P	6.63
1	ABM3B-8.000MHZ-B2-T / 535-9720-1-ND	Y1	8MHz 18pF / Abracon_ABM3B_5.0x3.2mm	0.76
3	AO3401A / C15127	Q101, Q201, Q301	AO3401A / SOT-23	0.285
1	AO4485 / C51499	Q1	AO4485 / SOIC-8_3.9x4.9mm_P1.27mm	0.2971
1	AP2112K-3.3TRG1 / AP2112K-3.3TRG1DICT-ND	U2	AP2112K-3.3 / SOT-23-5	0.28
3	BAT46W-E3-18 / C5557780	D102, D202, D302	BAT46W / D_SOD-123	0.3921
3	BSS138BK,215 / C282529	Q102, Q202, Q302	BSS138BK / SOT-23	0.2022
2	C0805C300J5GACTU / 399-C0805C300J5GACTUCT-ND	C25, C26	30pF / C_0805_2012Metric	0.66
4	C1005X7R1H104K050BB / 445-5932-1-ND	C61, C65, C66, C67	100nF / C_0402_1005Metric	0.4
3	CC0805JRNPO9BN102 / 311-1122-1-ND	C101, C201, C301	1nF / C_0805_2012Metric	0.45
2	CC0805KKX7R9BB105 / 311-1886-1-ND	C7, C23	1uF / C_0805_2012Metric	0.64
1	CC0805KRX7R9BB333 / 311-1374-1-ND	C69	33nF / C_0805_2012Metric	0.14
11	CL21B104KCFNNNE / C28233	C3, C6, C11, C20, C21, C22, C27, C40, C41, C50, C51	100nF / C_0805_2012Metric	0.4521
4	CL31A226KAHNNNE / C12891	C4, C5, C13, C12	22uF / C_1206_3216Metric	0.6836
7	CL32B106KBJNNWE / C3844168	C2, C60, C70, C62, C63, C64, C68	10uF / C_1210_3225Metric	3.4888
1	EEE-FK1H470P / PCE3815CT-ND	C1	47uF / CP_Elec_8x6.5	0.93

Qty	MPN / supplier part	References	Value / package	USD first tier
1	ESD2CAN24DBZRQ1 / 296-ESD2CAN24DBZRQ1CT-ND	D20	ESD2CAN24-Q1 / SOT-23	0.87
1	LM339LVDR / 296-LM339LVDRCT-ND	U7	LM339LV / SOIC-14_3.9x8.7mm_P1.27mm	0.42
1	LMR51420XDDCR / C5246146	U1	LMR51420XDDCR / TSOT-23-6	1.0292
1	LTST-C170KGKT / 160-1414-1-ND	D10	GREEN / LED_0805_2012Metric	0.19
3	M22-2010305 / 952-2274-ND	JP101, JP201, JP301	5V / 12V / PinHeader_1x03_P2.00mm_Vertical	0.66
1	MMSZ12T1G / MMSZ12T1GOSCT-ND	D3	12V ZENER / D_SOD-123	0.17
1	PT1206FR-070R1L / 311-0.1ANCT-ND	R64	0.1 / R_1206_Kelvin_Sense_Taps	0.35
3	RC0805FR-0720KL / 311-20.0KCRCT-ND	R103, R203, R303	20k / R_0805_2012Metric	0.3
3	RC0805FR-07220KL / 311-220KCRCT-ND	R101, R201, R301	220k / R_0805_2012Metric	0.33
1	RC0805FR-0724K9L / 311-24.9KCRCT-ND	R62	24.9k / R_0805_2012Metric	0.1
3	RC0805FR-072M2L / 311-2.20MCRCT-ND	R104, R204, R304	2.2M / R_0805_2012Metric	0.3
1	RC0805FR-0730K1L / 311-30.1KCRCT-ND	R66	30.1k / R_0805_2012Metric	0.1
1	RC0805FR-0759KL / 311-59.0KCRCT-ND	R10	59k / R_0805_2012Metric	0.1
1	RC0805FR-075K23L / 311-5.23KCRCT-ND	R51	5.23k / R_0805_2012Metric	0.1
1	RC0805JR-070RL / 311-0.0ARCT-ND	R20	0 / R_0805_2012Metric	0.1
3	RC1206FR-0710KL / 311-10.0KFRCT-ND	R109, R209, R309	10k / R_1206_3216Metric	0.33
1	RC1206FR-07120RL / 311-120FRCT-ND	R42	120 / R_1206_3216Metric	0.1
1	RT0805BRD07100KL / YAG1759CT-ND	R60	100k / R_0805_2012Metric	0.1
1	RT0805BRD07102KL / YAG4817CT-ND	R1	102k / R_0805_2012Metric	0.1
1	RT0805BRD0711K1L / 13-RT0805BRD0711K1LCT-ND	R61	11.1k / R_0805_2012Metric	0.1
1	RT0805BRD0713K7L / YAG4829CT-ND	R2	13.7k / R_0805_2012Metric	0.1
3	SMAJ30A / C148230	D101, D201, D301	SMAJ30A / D_SMA	0.4377

Qty	MPN / supplier part	References	Value / package	USD first tier
1	SMBJ22A-E3/52 / SMBJ22A-E3/52GICT-ND	D2	SMBJ22A / D_SMB	0.56
1	SRP5030TA-3R3M / SRP5030TA-3R3MCT-ND	L2	3.3uH / L_Bourns_SRP5030T	1.17
1	SRP7028A-100M / SRP7028A-100MCT-ND	L1	10uH / L_Bourns_SRP7028A_7.3x6.6mm	1.1
1	STM32F042K6T6 / 497-14647-ND	U5	STM32F042K6T6 / LQFP-32_7x7mm_P0.8mm	4.34
1	TCAN3413DR / C22433320	U6	TCAN3413DR / SOIC-8_3.9x4.9mm_P1.27mm	1.0227
2	TMK212AB7475KG-T / 587-2990-1-ND	C8, C24	4.7uF / C_0805_2012Metric	0.52
1	TPS2553DBVR-1 / C111738	U4	TPS2553DBVR-1 / SOT-23-6	0.2935
1	TPS552872QWRYQRQ1 / 296-TPS552872QWRYQRQ1CT-ND	U8	TPS552872-Q1 / TI_RYQ0021B_VQFN_HR_5x3mm	7.05